Online Retail Sales & Customer Analytics

Power BI Project — Findings, Insights & Business Recommendations

Project Overview

This project analyzes the UCI Online Retail transaction dataset to understand sales performance, product performance, customer behavior, returns, and international markets. The work covered Power Query cleaning, DAX measures, Power BI visualization, and business-oriented interpretation.

Analytical workflow: Raw Excel Dataset → Power Query → Data Cleaning → DAX → Analysis → Dashboard → Business Insights.

1. Data Cleaning & Preparation

- Created Revenue as Quantity × UnitPrice.
- Converted InvoiceDate into Date and created Year, Month Name and Month Number.
- Separated return/cancellation transactions from other negative transactions.
- Identified negative UnitPrice records associated with Adjust Bad Debt and excluded them from product analysis.
- Retained zero-price transactions while recognizing that they contribute zero revenue.
- Kept blank CustomerID rows for overall sales analysis, but excluded them from customer-level analysis.
- Created cleaned product descriptions and retained original fields.
- Excluded Postage, Manual, Adjust Bad Debt and Bank Charges from relevant product analyses.

2. Sales & Revenue Findings

- Overall revenue was approximately £9.77M.
- 2010 revenue was approximately £748,957; the dataset contains only December 2010.
- 2011 revenue was approximately £9.02M.
- The calculated 2010-to-2011 growth was 1104.46%, but this is not a full-year comparison because 2010 contains only December.
- November 2011 was the highest-revenue month in the monthly analysis.
- The UK generated approximately £8.21M, substantially more than other countries.
- Monthly peaks were strongly influenced by sales volume, not just price.

3. Product Performance Findings

- Postage generated substantial revenue but is a service item rather than a physical product, so it was excluded from physical-product rankings.
- Regency Cakestand 3 Tier was the highest-revenue physical product at approximately £164.8K.
- Other leading physical products included White Hanging Heart T-Light Holder (~£99.7K), Party Bunting (~£98.3K), and Jumbo Bag Red Retrospot (~£92.4K).
- Revenue is influenced by both quantity and unit price; high volume alone does not determine revenue.
- The Regency Cakestand analysis illustrated the interaction between price and quantity.

4. Customer Behavior Findings

- Approximately 4,373 customers had identifiable CustomerID.

- Customer 14646 was the highest-revenue customer at approximately £279.5K, with 74 sales orders, 196,719 units and 703 unique products.
- Customer 12748 had 210 orders and Customer 14911 had 201 orders.
- High purchase frequency does not necessarily mean high revenue.
- Average orders per customer were approximately 4.27.
- The top 5 identifiable customers contributed approximately 11.80% of customer-attributed revenue.

5. Customer Retention Findings

- 1,494 customers made exactly one sales order, approximately 34.17% of identifiable customers.
- One-time customers generated approximately £5.21M in customer-attributed revenue.
- Average revenue per one-time customer was approximately £348.80 in the report's displayed currency context.
- Average revenue per repeat customer was approximately £2,737.91.
- Repeat customers were associated with substantially higher average revenue per customer, although this does not establish causation.
- One-time customers are not necessarily low-value individually.

6. Returns & Cancellation Findings

- Return/cancellation transactions were separated from ordinary sales.
- Return Quantity was calculated as the absolute value of recorded negative quantities for return/cancellation transactions.
- Two complementary views were created: Top 10 Products by Return Quantity and Top 10 Products by Return Rate.
- Return Quantity measures returned-unit volume, while Return Rate measures return activity relative to recorded sales volume.
- A minimum sales-volume threshold of 100 units was used for the meaningful return-rate measure.
- Some products can show return-to-sales ratios above 100% because recorded return transactions can exceed recorded sales quantities for a product; this is not a literal probability.
- The dataset does not provide return reasons, so high return activity is an investigation signal rather than proof of product defects.

7. International Market Findings

- The UK is the dominant market in revenue and quantity.
- International markets show different relationships between revenue and sales volume.
- Some countries generate meaningful revenue without having the same volume as the largest markets, indicating differences in product mix and selling value.
- Separating the UK makes smaller international markets easier to compare visually.

8. Key Business Insights

- Revenue performance is strongly concentrated in the UK, while international markets show different revenue-to-volume patterns.
- Regency Cakestand 3 Tier is the highest-revenue physical product, demonstrating the combined effect of price and quantity.
- Customer purchase frequency and monetary value are separate dimensions.

- About one-third of identifiable customers made only one purchase, while repeat customers had much higher average revenue per customer.
- Return quantity and return rate provide different perspectives on product performance.
- International markets should be evaluated using both revenue and volume rather than volume alone.

9. Business Recommendations

- Improve retention: develop targeted follow-up strategies for one-time customers to encourage repeat purchases.
- Segment customers using both purchase frequency and monetary value.
- Investigate products with unusually high return activity using product descriptions, pricing, order handling and customer feedback where available.
- Evaluate international markets using revenue, quantity, order frequency and product mix.
- Use both quantity and revenue when identifying strong products because high-volume and high-value products can require different strategies.

10. Important Data Limitations

- The dataset contains only December 2010 plus the full 2011 period, so the 2010-to-2011 growth rate is not a full-year comparison.
- The dataset has no product cost field, so actual profit and profit margin cannot be calculated reliably.
- Blank CustomerID transactions cannot be attributed to individual customers.
- Postage, accounting adjustments and manual entries were excluded from relevant product analyses.
- Return reasons are not available, so return analysis identifies patterns for investigation rather than explaining causes.
- The original UCI Online Retail dataset is commonly represented in GBP; currency labels in the Power BI report should be kept consistent.

11. Interview-Ready Project Summary

“I built an end-to-end Power BI sales and customer analytics project using the UCI Online Retail dataset. I cleaned and transformed the transaction data in Power Query, created DAX measures for revenue, orders, customers, ranking and return metrics, and developed dashboards covering sales performance, product performance, customer behavior, retention, returns and international markets. The analysis showed differences between purchase frequency and customer value, identified a significant one-time customer segment, highlighted high-return products for further investigation, and showed that revenue and sales volume vary across international markets. I then translated those findings into data-backed business recommendations.”